文献汇报

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Object Detection in Video with Spatiotemporal Sampling Networks

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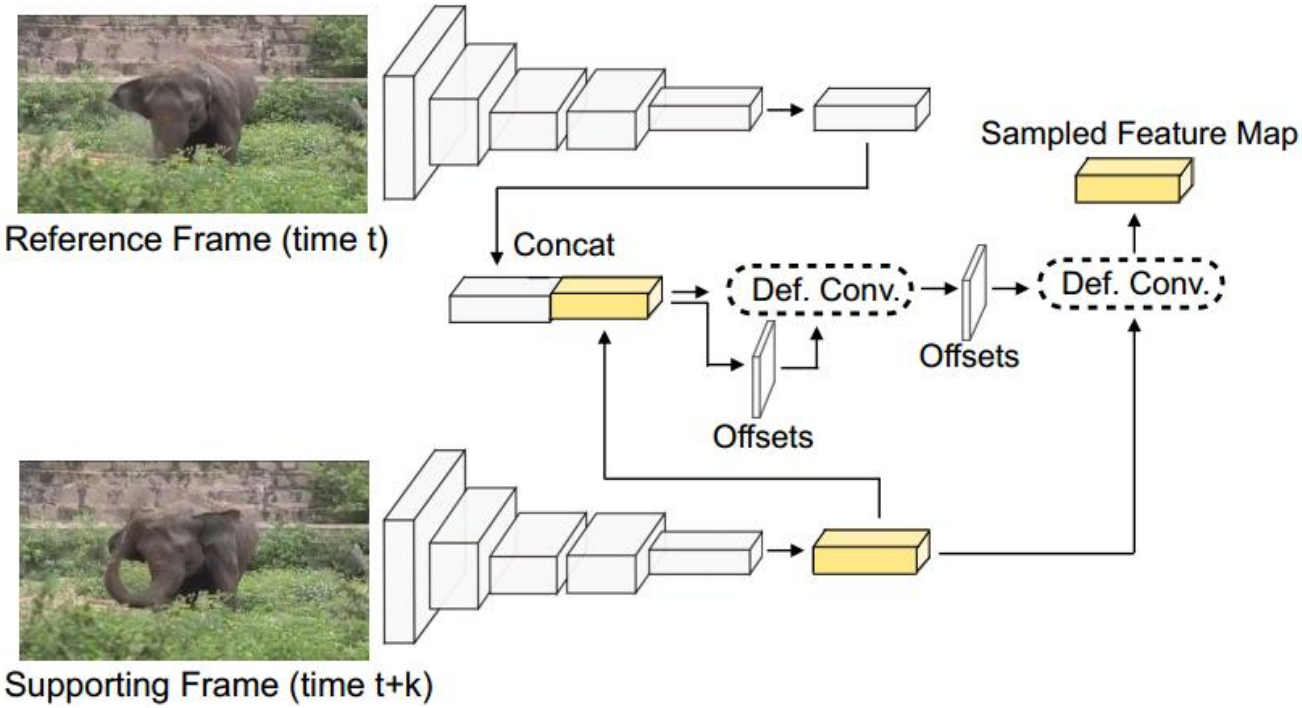
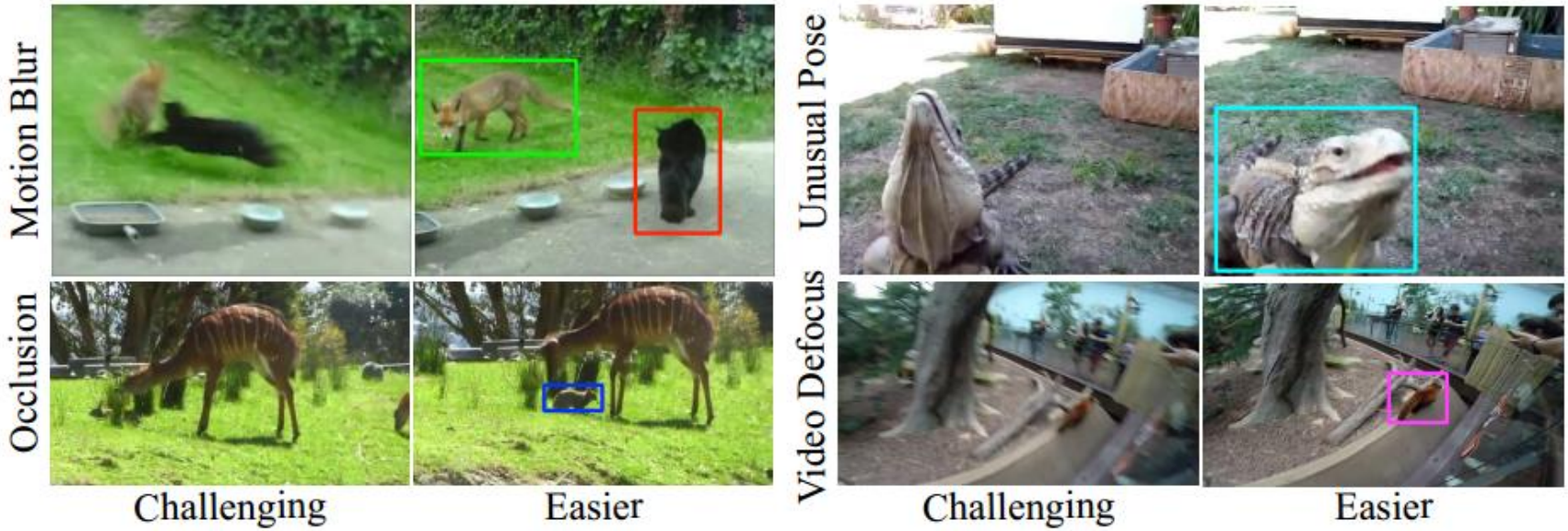


Method

与FGFA十分相似，思想都是单帧不够，多帧来凑，利用相邻帧的特征来对当前帧进行增强，以此来达到更好的检测效果。不同点在于FGFA是利用光流来对两张图片的特征进行对齐，这篇文章采用的是可变形卷积来做特征对齐。

将可变形卷积的卷积结果作为对齐后的特征，之后按照FGFA一样的方式进行加权求和，得到增强后的特征。同FGFA相比，这篇文章利用可变形卷积的方法。



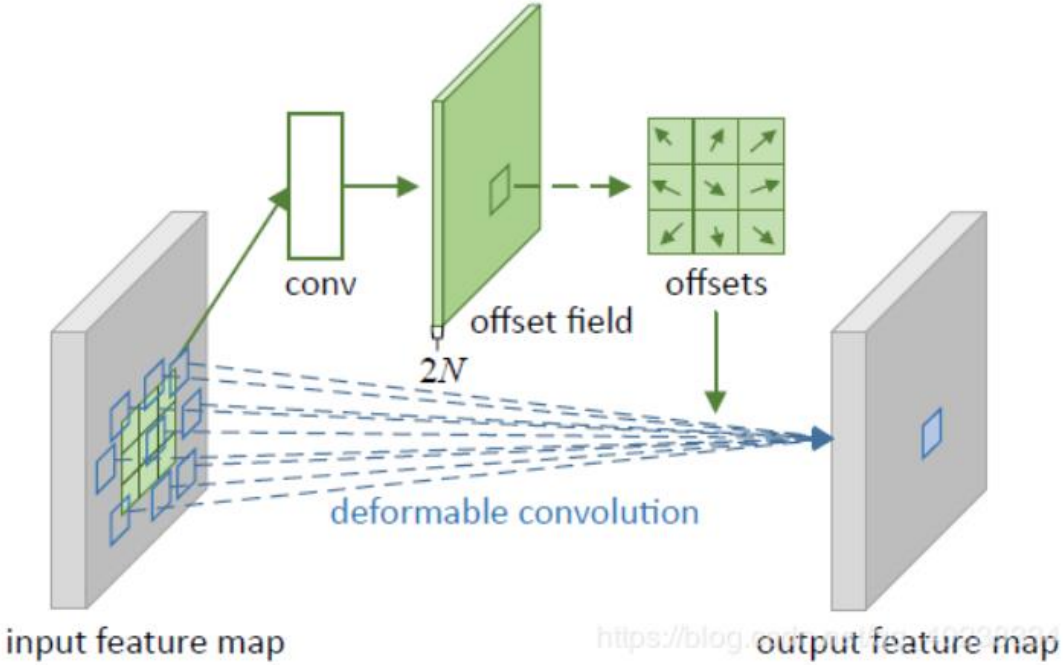
Deformable Convolution

普通卷积

$$y(p_0) = \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0 + p_n),$$

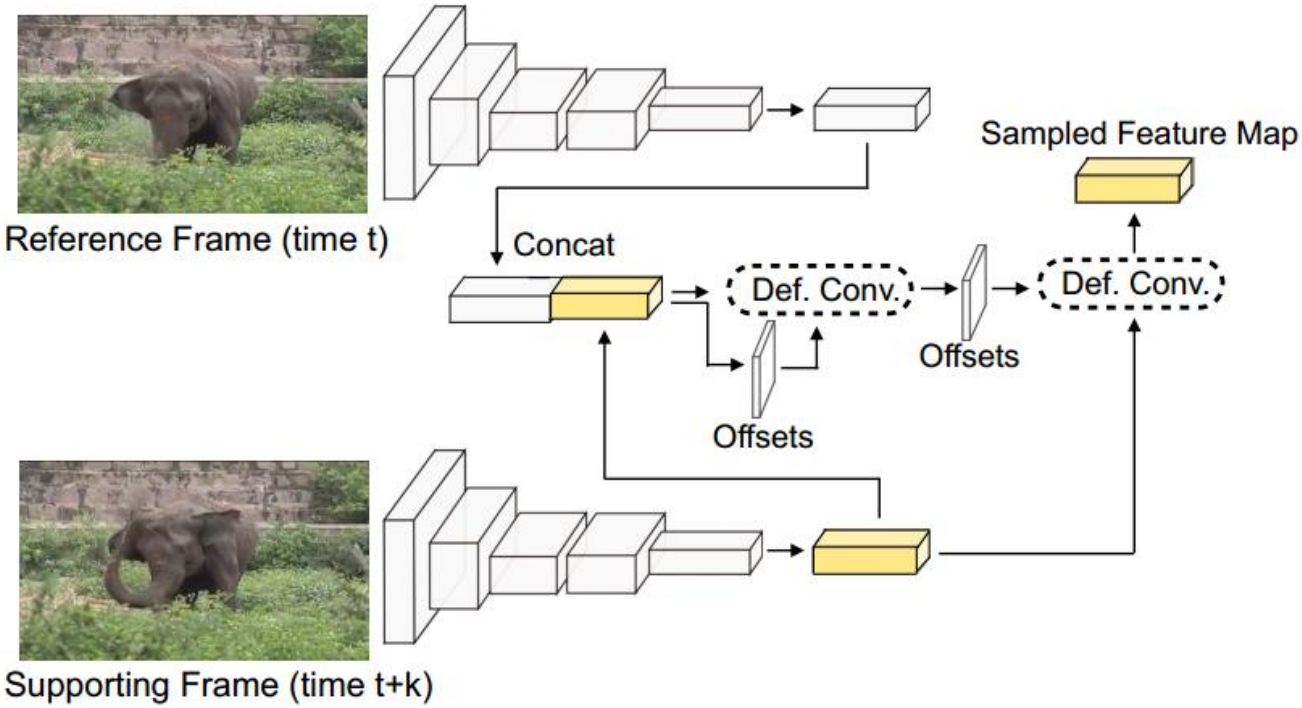
可变性卷积

$$y(p_0) = \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0 + p_n + \Delta p_n)$$



细节

特征提取网络使用基于可变性卷积的ResNet网络，时空特征采样使用四个可变性卷积，下图中只画出了两个。后续检测网络使用R-FCN。



特征聚合

从Deformable Convolution得到的前后相邻帧（共2K+1个）的采样特征图后，根据FGFA里面的余弦相似度的方法计算权重值，对这2K+1个采样特征图加权平均后得到当前帧最终的特征图

$$g_t^{agg}(p) = \sum_{k=-K}^{K} w_{t,t+k}(p)\ g_{t,t+k}^{(4)}(p)$$

$$w_{t,t+k}(p) = \exp\left(\frac{S(g_{t,t}^{(4)})(p) \cdot S(g_{t,t+k}^{(4)})(p)}{|S(g_{t,t}^{(4)})(p)||S(g_{t,t+k}^{(4)})(p)|}\right)$$

Experiments

Inference. During inference, we use $T = 27$, meaning that we consider $K = 13$ supporting frames before and after the reference frame. To avoid GPU memory issues, we first extract features from the backbone network for each image individually, and then cache these features in the memory. Afterwards, we feed all these features into our spatiotemporal sampling block. At the end, standard NMS with a threshold of 0.3 is applied to refine the detections. To handle the first and the last $K = 13$ frames in the video —two boundary cases that require sampling the neighboring frames beyond the video start and end, we pad the start of a video with K copies of the first frame, and the end of a video with K copies of the last frame.

	Methods					
	D&T [27]	Our SSN	FGFA [24]	Our STSN	D&T+ [27]	Our STSN+
No FlowNet?	✓	-	✗	✓	✓	✓
Not Using Flow Data?	✓	-	✗	✓	✓	✓
No Temporal Post-Processing?	✓	-	✓	✓	✗	✗
mAP@0.5	75.8	76.0	78.8	78.9	79.8	**80.4**